CAS SciFinder

Task History

Initiating Search

November 6, 2025, 4:18 PM

Search:

CAS Lexicon Search:

Knoevenagel reaction - Preferred Concept

Search Tasks

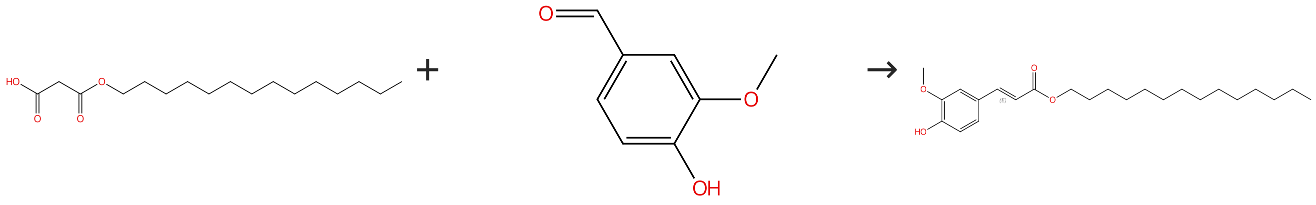
Task	Result Type	View
Returned Reference Results from CAS Lexicon (9,030)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,468) Filtered By: Yield: 90-100%, 80-89% Document Type: Journal Language: English Publication Name: European Journal of Medicinal Chemistry, Bioorganic & Medicinal Chemistry Letters, Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry, Medicinal Chemistry Research, Journal of Chemical and Pharmaceutical Research, Organic and Medicinal Chemistry Letters, Organic Preparations and Procedures International	Reactions	View Results

CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (4) | Suppliers (145) | Double bond geometry shown | Supplier (1)

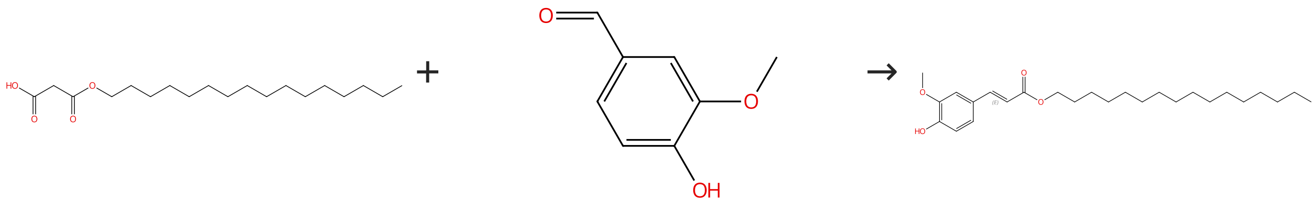
31-333-CAS-6108306	Steps: **1** Yield: **100%**
1.1	**Reagents:** Pyridine **Catalysts:** β-Alanine **Solvents:** Pyridine; 2 w, rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; 0 °C
Experimental Protocols	

Synthesis and antioxidant activity of long chain alkyl hydroxycinnamates

By: Menezes, Jose C. J. M. D. S.; et al

European Journal of Medicinal Chemistry (2011), 46(2), 773-777.

Scheme 2 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (9) | Suppliers (145) | Double bond geometry shown | Suppliers (14)

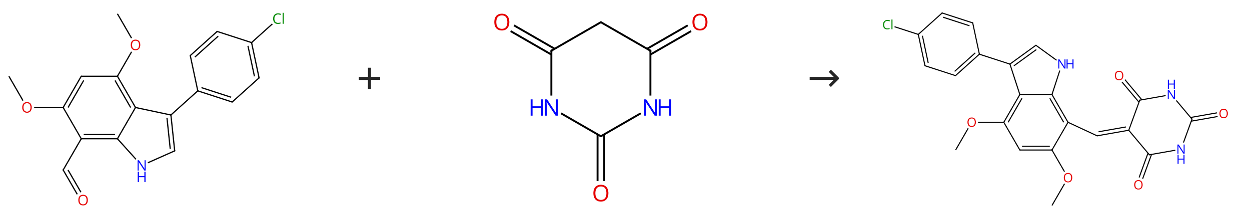
31-333-CAS-14625827	Steps: **1** Yield: **100%**
1.1	**Reagents:** Pyridine **Catalysts:** β-Alanine **Solvents:** Pyridine; 2 w, rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; 0 °C
Experimental Protocols	

Synthesis and antioxidant activity of long chain alkyl hydroxycinnamates

By: Menezes, Jose C. J. M. D. S.; et al

European Journal of Medicinal Chemistry (2011), 46(2), 773-777.

Scheme 3 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (2) | Suppliers (81) | Supplier (1)

31-331-CAS-17945804	Steps: **1** Yield: **100%**
1.1	**Reagents:** Acetic anhydride; 2 h, 90 °C
Experimental Protocols	

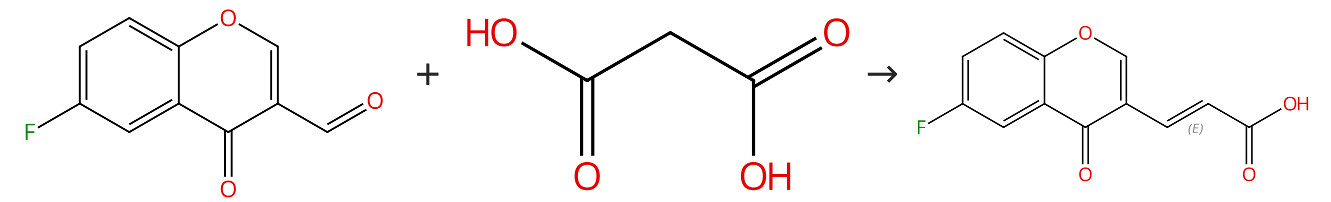
Small molecule inhibitors of bacterial transcription complex formation

By: Wenholz, Daniel S.; et al

Bioorganic & Medicinal Chemistry Letters (2017), 27(18), 4302-4308.

Scheme 4 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (77)

Suppliers (87)

Double bond geometry shown

Suppliers (11)

31-333-CAS-16671497	Steps: **1** Yield: **100%**
1.1	**Reagents:** Pyridine; 1.5 h, 110 °C
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; neutralized
Experimental Protocols	

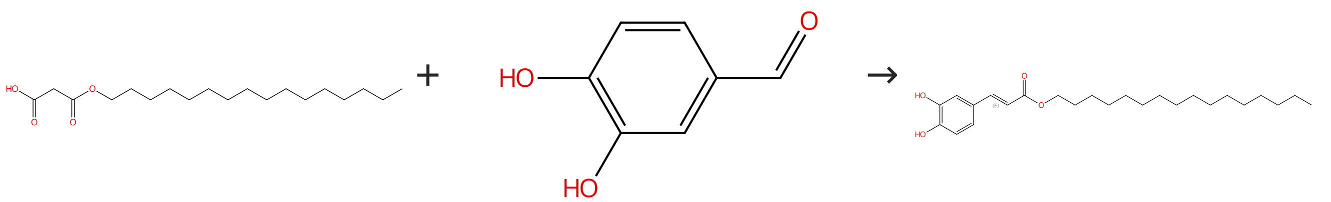
Synthesis and characterization of new N-alkylated pyridine-2 (1H)-ones

By: Sharmaa, Atul Kumar; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2016), 55B(4), 492-500.

Scheme 5 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (9)

Suppliers (112)

Double bond geometry shown

Suppliers (2)

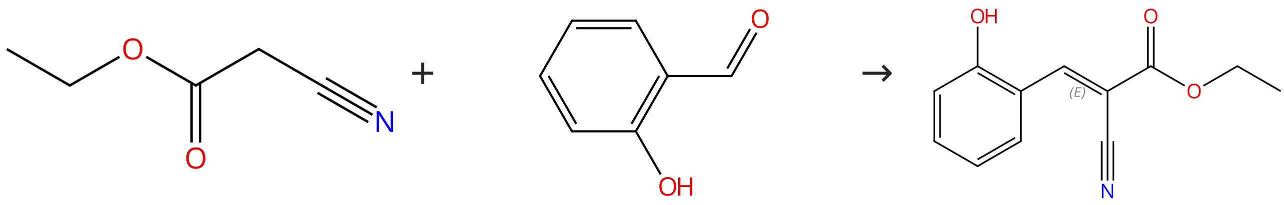
31-333-CAS-10367697	Steps: **1** Yield: **100%**
1.1	**Reagents:** Pyridine **Catalysts:** β-Alanine **Solvents:** Pyridine; 2 w, rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; 0 °C
Experimental Protocols	

Synthesis and antioxidant activity of long chain alkyl hydroxycinnamates

By: Menezes, Jose C. J. M. D. S.; et al

European Journal of Medicinal Chemistry (2011), 46(2), 773-777.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (70)

Suppliers (95)

Double bond geometry shown

Suppliers (10)

31-331-CAS-8246651 Steps: **1** Yield: **100%**

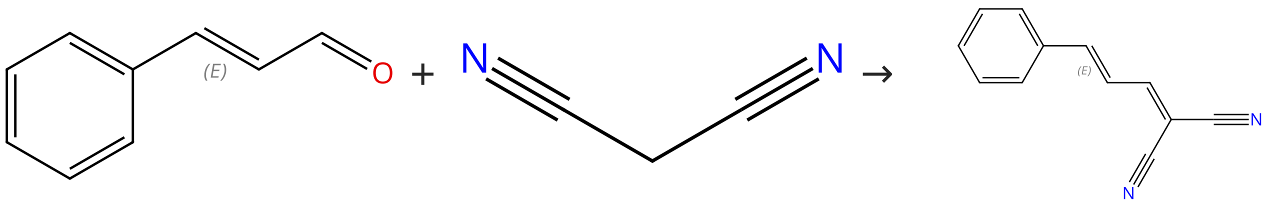
1.1 **Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine); 6 min, rt

Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions

By: Krishnan, G. Rajesh; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown

Suppliers (109)

Suppliers (63)

Double bond geometry shown

Suppliers (7)

31-333-CAS-6769163 Steps: **1** Yield: **100%**

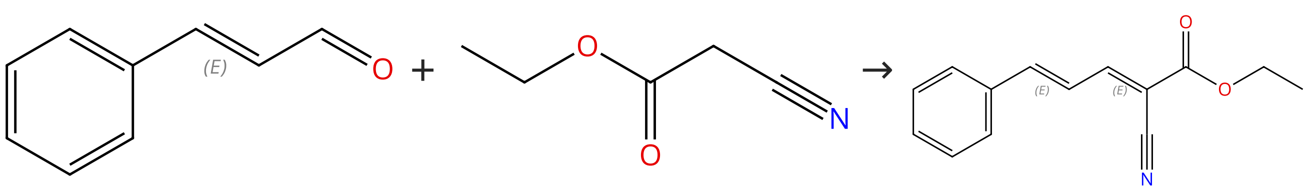
1.1 **Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine)
Solvents: Water; 20 min, rt

Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions

By: Krishnan, G. Rajesh; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown

Suppliers (109)

Suppliers (70)

Double bond geometry shown

Suppliers (5)

31-331-CAS-1534708 Steps: **1** Yield: **100%**

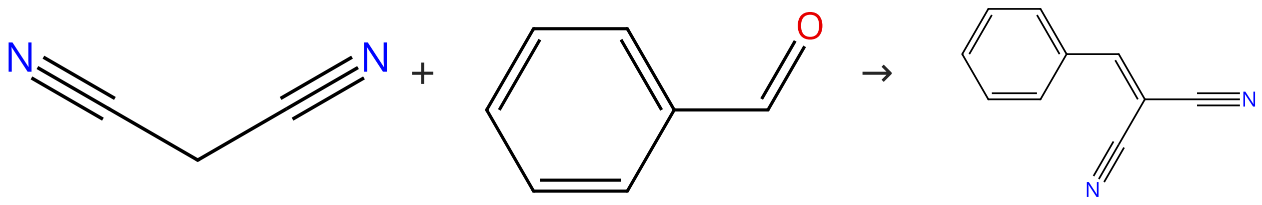
1.1 **Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine); 5 min, rt

Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions

By: Krishnan, G. Rajesh; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

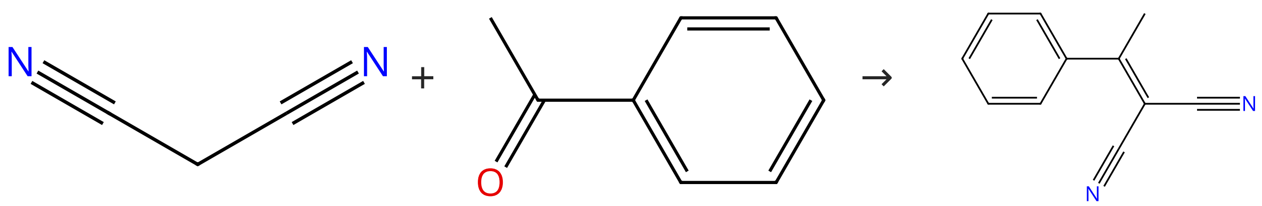
Scheme 9 (4 Reactions) Steps: **1** Yield: **98-100%**



Suppliers (63) Suppliers (69) Suppliers (73)

31-333-CAS-9116897 Steps: **1** Yield: **100%** 1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1) **Solvents:** Water; 1 min, rt Experimental Protocols	**A simple and efficient procedure for the Knoevenagel condensation catalyzed by [MeHMTA]BF_4 ionic liquid** By: Keithellakpam, Sanjoy; et al Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.
31-333-CAS-716912 Steps: **1** Yield: **100%** 1.1 **Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine) **Solvents:** Water; 30 min, rt	**Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions** By: Krishnan, G. Rajesh; et al Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.
31-333-CAS-13103704 Steps: **1** Yield: **99%** 1.1 **Catalysts:** 1-Octanesulfonic acid, 1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,8-heptadecafluoro-, ytterbium(3+) salt (3:1) **Solvents:** Toluene, Perfluorodecalin; 2 h, 80 °C Experimental Protocols	**Ytterbium perfluorooctanesulfonate-catalyzed Knoevenagel condensation in a fluorous biphasic system** By: Yi, Wen-Bin; et al Organic Preparations and Procedures International (2006), 39(1), 71-75.
31-333-CAS-2435834 Steps: **1** Yield: **98%** 1.1 **Reagents:** Ammonium sulfamate; 10 s	**Microwave-enhanced Knoevenagel condensation catalyzed by $NH_2SO_3NH_4$** By: Mogilaiah, K.; et al Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2010), 49B(3), 390-393.

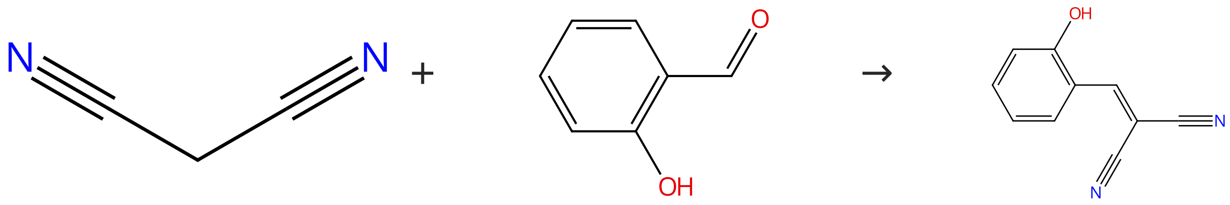
Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (63) Suppliers (105) Suppliers (50)

31-333-CAS-10700411 Steps: **1** Yield: **100%** 1.1 **Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine); 7 min, rt	**Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions** By: Krishnan, G. Rajesh; et al Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (63) Suppliers (95) Suppliers (15)

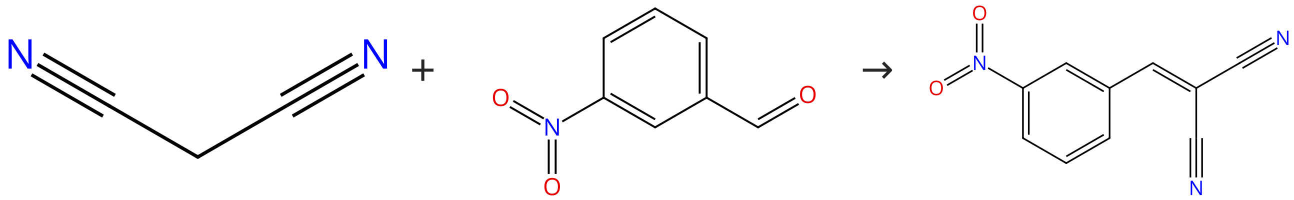
31-333-CAS-9515383	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine); 6 min, rt

Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions

By: Krishnan, G. Rajesh; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (63) Suppliers (90) Suppliers (27)

31-333-CAS-1965046	Steps: **1** Yield: **100%**
1.1	**Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1) **Solvents:** Water; 1 min, rt

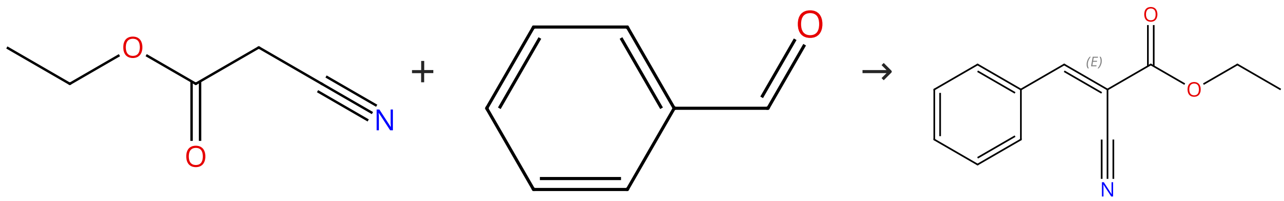
Experimental Protocols

A simple and efficient procedure for the Knoevenagel condensation catalyzed by [MeHMTA]BF_4 ionic liquid

By: Keithellakpam, Sanjoy; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.

Scheme 13 (4 Reactions) Steps: **1** Yield: **98-100%**



Suppliers (70) Suppliers (69)

Double bond geometry shown

Suppliers (43)

31-331-CAS-8856746	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine); 6 min, rt

Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions

By: Krishnan, G. Rajesh; et al

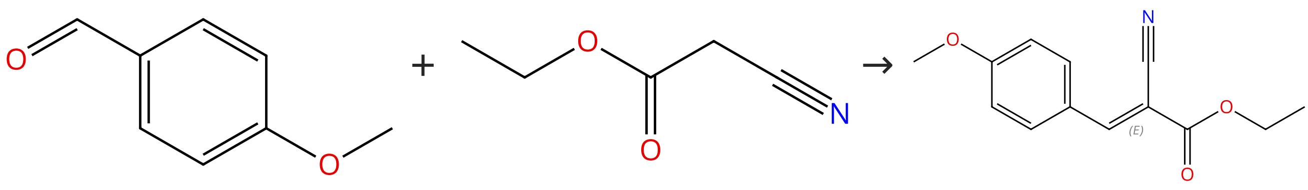
Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

31-614-CAS-24070277 Steps: **1** Yield: **98%**	**Ammonium chloride catalyzed Knoevenagel condensation in PEG-400 as ecofriendly solvent**
1.1 **Catalysts:** Ammonium chloride **Solvents:** Polyethylene glycol; 60 min, rt 1.2 **Solvents:** Water; 5 °C	By: Waghmare, Smita R.
Experimental Protocols	Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2021), 60B(6), 849-855.

31-331-CAS-8509756 Steps: **1** Yield: **98%**	**A simple and efficient procedure for the Knoevenagel condensation catalyzed by [MeHMTA]BF_4 ionic liquid**
1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1) **Solvents:** Water; 5 min, rt	By: Keithellakpam, Sanjoy; et al
Experimental Protocols	Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.

31-331-CAS-14841467 Steps: **1** Yield: **98%**	**Knoevenagel reactions catalyzed by ionic liquids**
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium acetate; 60 min, 25 °C	By: Hu, Xiaomei; et al
	Journal of Chemical and Pharmaceutical Research (2014), 6(4), 864-868, 5 pp..

Scheme 14 (1 Reaction) Steps: **1** Yield: **100%**



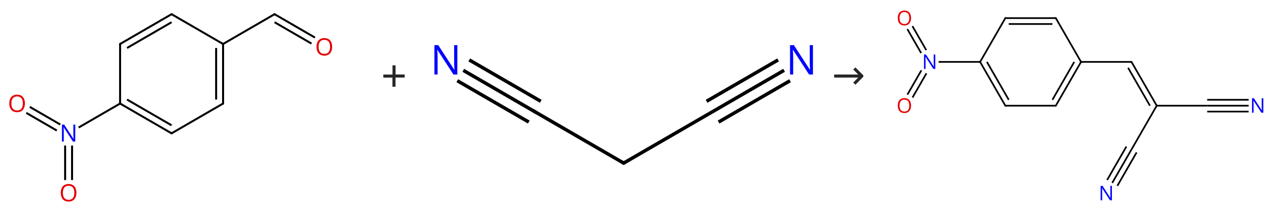
Suppliers (128) Suppliers (70)

Double bond geometry shown

Suppliers (45)

31-331-CAS-15238312 Steps: **1** Yield: **100%**	**Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions**
1.1 **Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine) **Solvents:** Water; 30 min, rt	By: Krishnan, G. Rajesh; et al
	Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

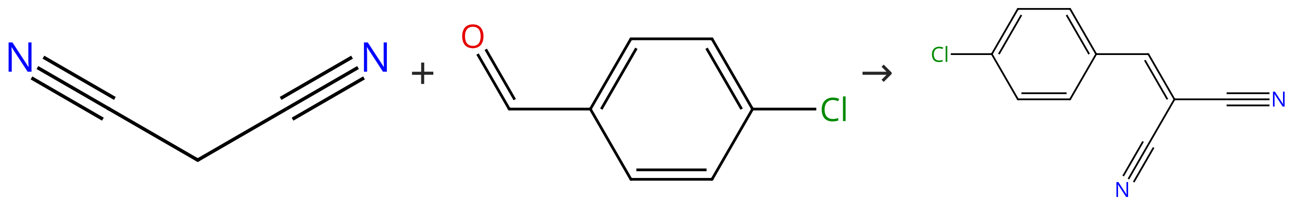
Scheme 15 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (92) Suppliers (63) Suppliers (53)

31-333-CAS-10587824 Steps: **1** Yield: **100%**	**A simple and efficient procedure for the Knoevenagel condensation catalyzed by [MeHMTA]BF_4 ionic liquid**
1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1) **Solvents:** Water; 1 min, rt	By: Keithellakpam, Sanjoy; et al
Experimental Protocols	Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.

Scheme 16 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (63) Suppliers (99) Suppliers (47)

31-333-CAS-1137281 Steps: **1** Yield: **100%**

1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1)
Solvents: Water; 2 min, rt

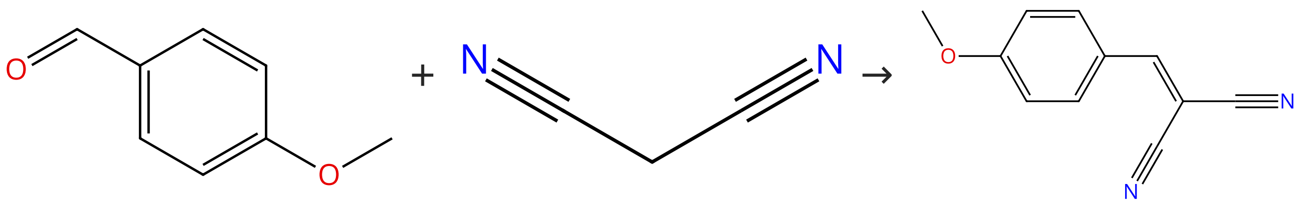
Experimental Protocols

A simple and efficient procedure for the Knoevenagel condensation catalyzed by [MeHMTA]BF_4 ionic liquid

By: Keithellakpam, Sanjoy; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.

Scheme 17 (2 Reactions) Steps: **1** Yield: **99-100%**



Suppliers (128) Suppliers (63) Suppliers (72)

31-333-CAS-11382607 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Poly(vinyl chloride) (reaction product with ethylene diamine)
Solvents: Water; 30 min, rt

Aminated poly(vinyl chloride), an efficient green catalyst for Knoevenagel condensation reactions

By: Krishnan, G. Rajesh; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(3), 428-431.

31-333-CAS-15184279 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1)
Solvents: Water; 2 min, rt

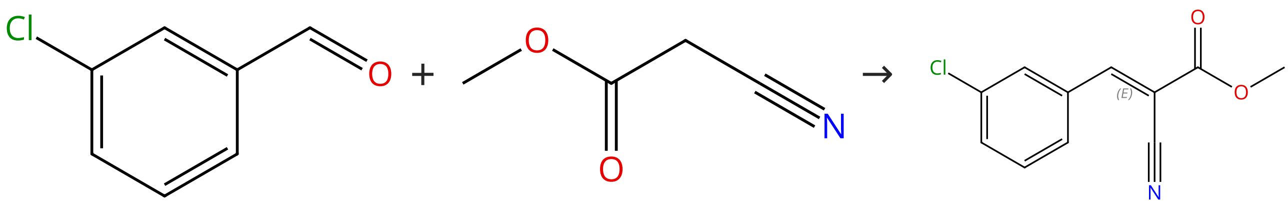
Experimental Protocols

A simple and efficient procedure for the Knoevenagel condensation catalyzed by [MeHMTA]BF_4 ionic liquid

By: Keithellakpam, Sanjoy; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.

Scheme 18 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (89) Suppliers (78)

Double bond geometry shown

Suppliers (3)

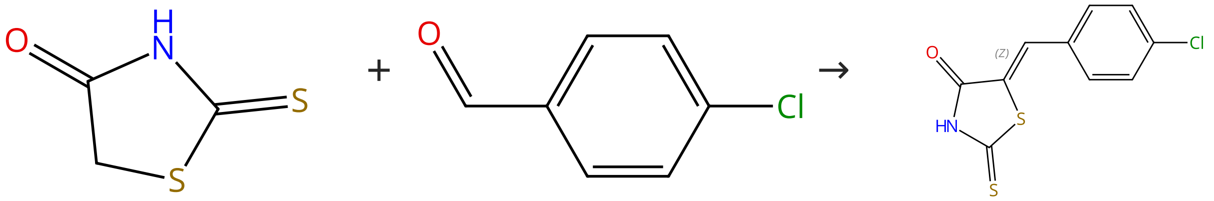
31-331-CAS-4760060	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 1-Butyl-3-methylimidazolium trifluoroacetate; 5 min, 25 °C

Knoevenagel reactions catalyzed by ionic liquids

By: Hu, Xiaomei; et al

Journal of Chemical and Pharmaceutical Research (2014), 6(4), 864-868, 5 pp..

Scheme 19 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (91) Suppliers (99)

Double bond geometry shown

Suppliers (14)

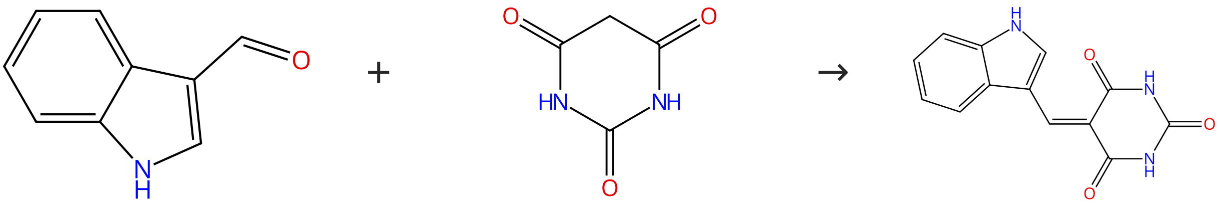
31-331-CAS-6975260	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 1,1,3,3-Tetramethylguanidinium lactate; 10 min, 80 °C
Experimental Protocols	

Ultrasound-assisted synthesis of 2,4-thiazolidinedione and rhodanine derivatives catalyzed by task-specific ionic liquid: [TMG][Lac]

By: Suresh; et al

Organic and Medicinal Chemistry Letters (2013), 3(1), 2/1-2/6.

Scheme 20 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (116) Suppliers (81) Suppliers (19)

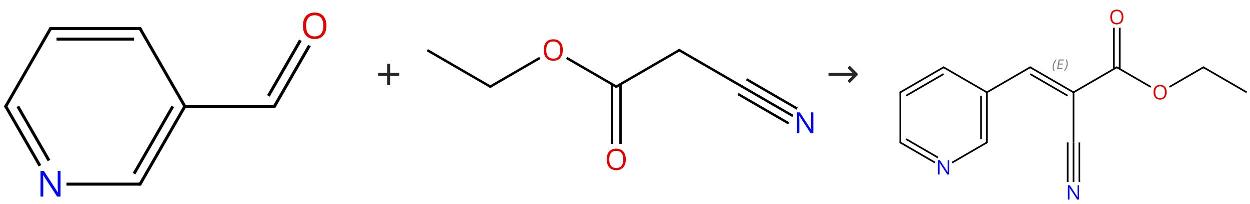
31-331-CAS-11697389	Steps: **1** Yield: **99%**
1.1	1 - 5 min

Design, synthesis and anticancer activities of hybrids of indole and barbituric acids-Identification of highly promising leads

By: Singh, Palwinder; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(11), 3054-3058.

Scheme 21 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (92) Suppliers (70)

Double bond geometry shown

Suppliers (26)

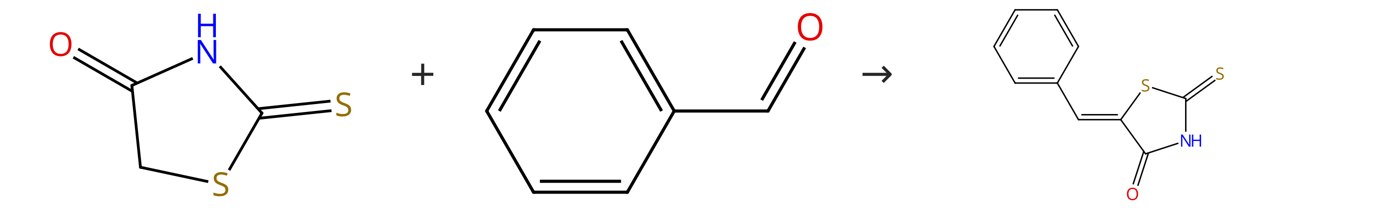
31-331-CAS-4583519	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 1-Octanesulfonic acid, 1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,8-heptadecafluoro-, ytterbium(3+) salt (3:1) **Solvents:** Toluene, Perfluorodecalin; 3 h, 80 °C
Experimental Protocols	

Ytterbium perfluorooctanesulfonate-catalyzed Knoevenagel condensation in a fluorous biphasic system

By: Yi, Wen-Bin; et al

Organic Preparations and Procedures International (2006), 39(1), 71-75.

Scheme 22 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (91) | Suppliers (69) | Suppliers (12)

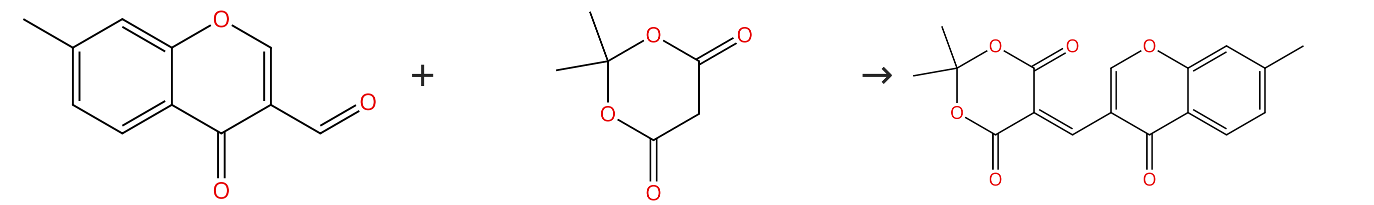
31-331-CAS-12471160	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Urea; 5 min, 110 °C

Urea/thiourea catalyzed, solvent-free synthesis of 5-arylidenethiazolidine-2,4-diones and 5-arylidene-2-thioxothiazolidin-4-ones

By: Shah, Sakshi; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(17), 5388-5391.

Scheme 23 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (23) | Suppliers (101)

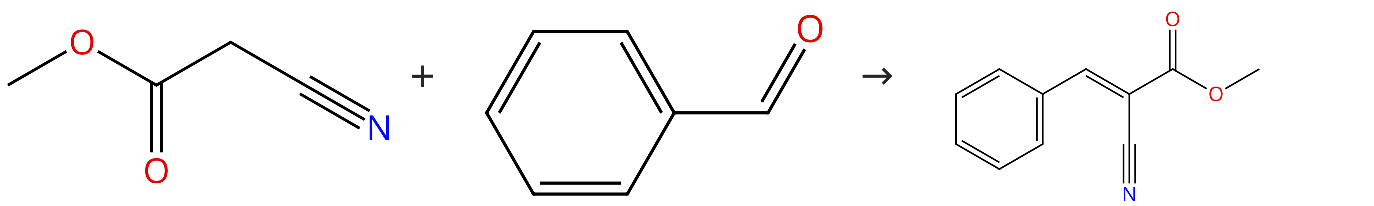
31-331-CAS-3093152	Steps: **1** Yield: **99%**
1.1	1 min, heated

Microwave-induced, solvent-free Knoevenagel condensation of 4-oxo-(4H)-1-benzopyran-3-carbaldehyde with Meldrum's acid using alumina support

By: Shindalkar, S. S.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2006), 45B(11), 2571-2573.

Scheme 24 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (78) | Suppliers (69) | Suppliers (36)

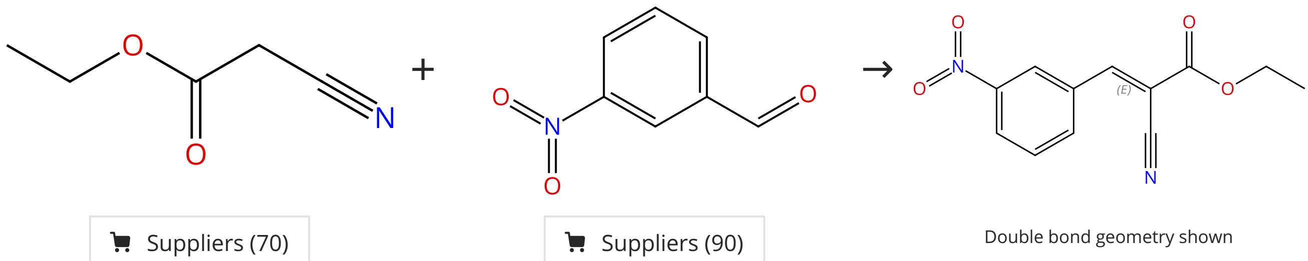
31-331-CAS-23057215	Steps: **1** Yield: **99%**
1.1 **Solvents:** Water; 3 min, 25 °C	
Experimental Protocols	

Facile Method for the Synthesis of Cyanoac rylates by Knoeve nagel Condensation

By: Mabaso, Thabile; et al

Organic Preparations and Procedures International (2021), 53(1), 18-24.

Scheme 25 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (70) Suppliers (90) Suppliers (12)

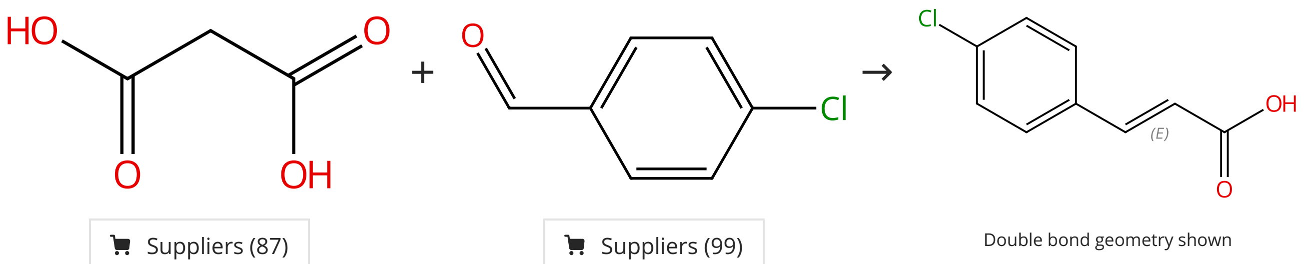
31-331-CAS-1630057	Steps: **1** Yield: **99%**
1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo$[3.3.1.1^{3,7}]$decane, 1-methyl-, tetrafluoroborate(1-) (1:1) **Solvents:** Water; 10 min, rt	
Experimental Protocols	

A simple and efficient procedure for the Knoeve nagel conden sation catalyzed by [Me HMTA]BF_4 ionic liquid

By: Keithellakpam, Sanjoy; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.

Scheme 26 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (87) Suppliers (99) Suppliers (56)

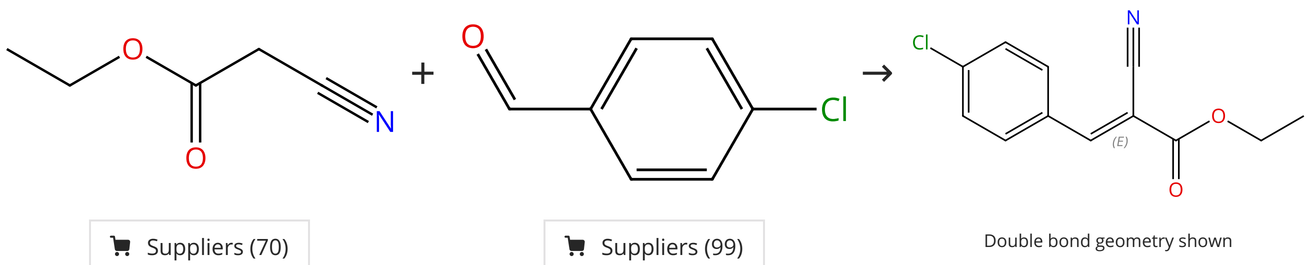
31-333-CAS-6385561	Steps: **1** Yield: **99%**
1.1 **Catalysts:** 1-Octanesulfonic acid, 1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,8-heptadecafluoro-, ytterbium(3+) salt (3:1) **Solvents:** Toluene, Perfluorodecalin; 8 h, 80 °C	
Experimental Protocols	

Ytterbium perfluorooctanesulfonate-catalyzed Knoevenagel condensation in a fluorous biphasic system

By: Yi, Wen-Bin; et al

Organic Preparations and Procedures International (2006), 39(1), 71-75.

Scheme 27 (3 Reactions) Steps: **1** Yield: **98-99%**



Suppliers (70) Suppliers (99) Suppliers (49)

31-331-CAS-11277228	Steps: **1** Yield: **99%**
1.1 **Catalysts:** 1-Octanesulfonic acid, 1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,8-heptadecafluoro-, ytterbium(3+) salt (3:1) **Solvents:** Toluene, Perfluorodecalin; 8 h, 80 °C	
Experimental Protocols	

Ytterbium perfluorooctanesulfonate-catalyzed Knoevenagel condensation in a fluorous biphasic system

By: Yi, Wen-Bin; et al

Organic Preparations and Procedures International (2006), 39(1), 71-75.

31-331-CAS-15357042	Steps: **1** Yield: **98%**
1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1) **Solvents:** Water; 10 min, rt	
Experimental Protocols	

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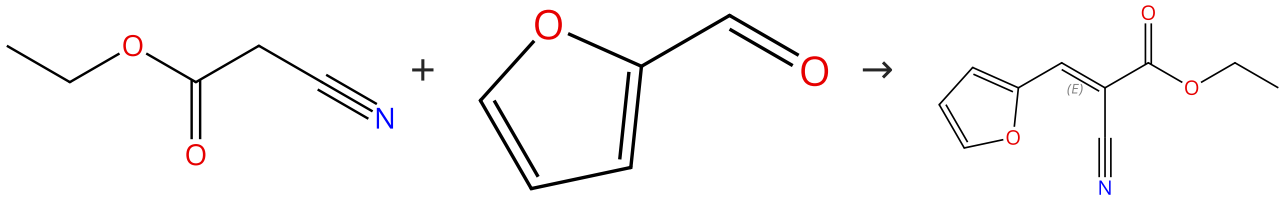
31-331-CAS-7844755	Steps: **1** Yield: **98%**
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium trifluoroacetate; 10 min, 25 °C	

Knoevenagel reactions catalyzed by ionic liquids

By: Hu, Xiaomei; et al

Journal of Chemical and Pharmaceutical Research (2014), 6(4), 864-868, 5 pp..

Scheme 28 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (70) Suppliers (81)

Double bond geometry shown

Suppliers (20)

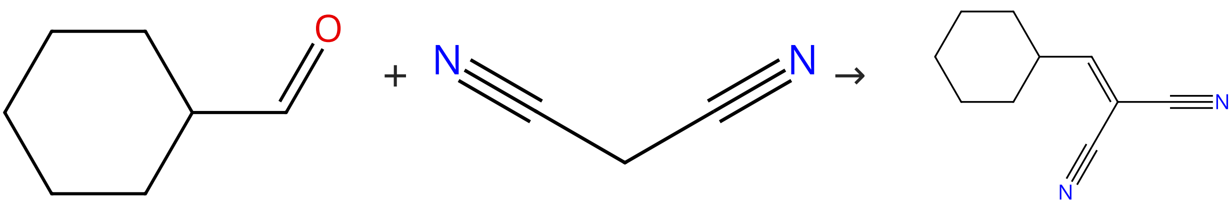
31-331-CAS-2453678	Steps: **1** Yield: **99%**
1.1 **Catalysts:** 1-Octanesulfonic acid, 1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,8-heptadecafluoro-, ytterbium(3+) salt (3:1) **Solvents:** Toluene, Perfluorodecalin; 3 h, 80 °C	
Experimental Protocols	

Ytterbium perfluorooctanesulfonate-catalyzed Knoevenagel condensation in a fluorous biphasic system

By: Yi, Wen-Bin; et al

Organic Preparations and Procedures International (2006), 39(1), 71-75.

Scheme 29 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (81) Suppliers (63) Suppliers (27)

31-333-CAS-591001	Steps: **1** Yield: **99%**
1.1 **Catalysts:** 3,5,7-Triaza-1-azoniatricyclo[3.3.1.1^{3,7}]decane, 1-methyl-, tetrafluoroborate(1-) (1:1) **Solvents:** Water; 5 min, rt	
Experimental Protocols	

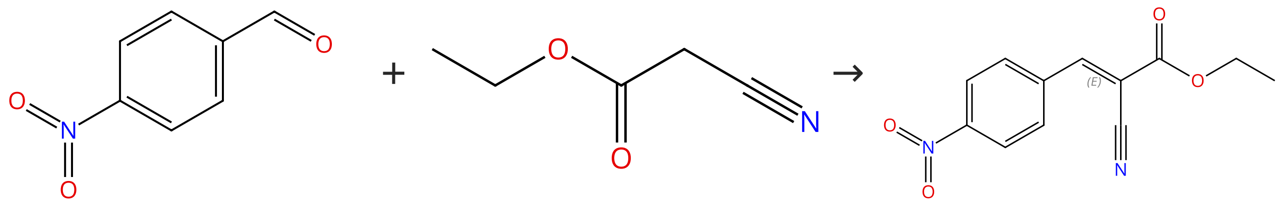
A simple and efficient procedure for the Knoeve nagel conden sation catalyzed by [Me HMTA]BF_4 ionic liquid

By: Keithellakpam, Sanjoy; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2015), 54B(9), 1157-1161.

Scheme 30 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (92)

Suppliers (70)

Double bond geometry shown

Suppliers (34)

31-331-CAS-10329

Steps: **1** Yield: **99%**

1.1 **Catalysts:** 1-Octanesulfonic acid, 1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,8-heptadecafluoro-, ytterbium(3+) salt (3:1)
Solvents: Toluene, Perfluorodecalin; 6 h, 80 °C

Experimental Protocols

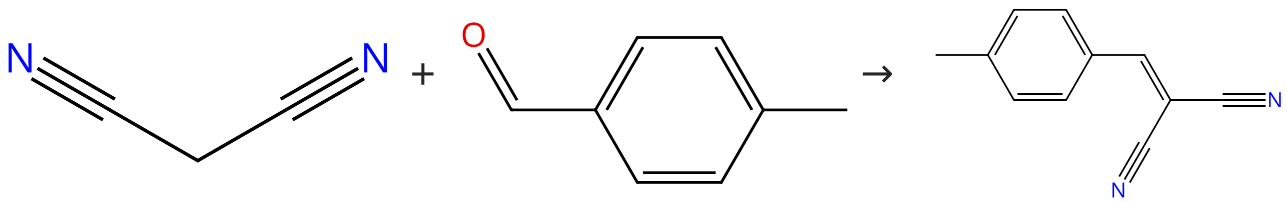
Ytterbium perfluorooctanesulfonate-catalyzed Knoevenagel condensation in a fluorous biphasic system

By: Yi, Wen-Bin; et al

Organic Preparations and Procedures International (2006), 39(1), 71-75.

Scheme 31 (2 Reactions)

Steps: **1** Yield: **98-99%**



Suppliers (63)

Suppliers (104)

Suppliers (58)

31-333-CAS-7157214

Steps: **1** Yield: **99%**

1.1 **Reagents:** Ammonium sulfamate; 10 s

Microwave-enhanced Knoevenagel condensation catalyzed by $NH_2SO_3NH_4$

By: Mogilaiah, K.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2010), 49B(3), 390-393.

31-333-CAS-186834

Steps: **1** Yield: **98%**

1.1 **Catalysts:** Ammonium fluoride; 30 s, heated

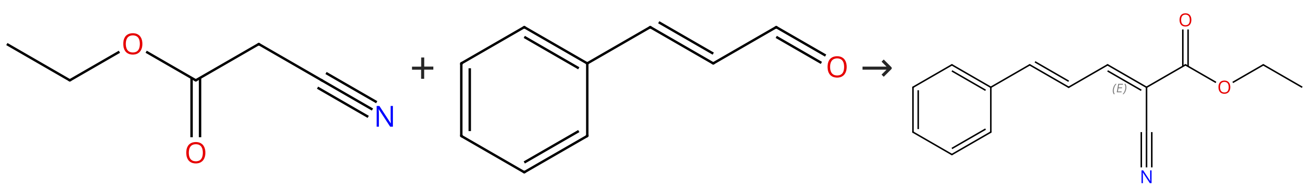
Ammonium fluoride as an inexpensive catalyst for Knoeve nagel condensation in solvent-free conditions under microwave irradiation

By: Mogilaiah, K.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2013), 52B(5), 694-697.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **98%**



Suppliers (70)

Suppliers (52)

E/Z labels describe double bond geometry

Supplier (1)

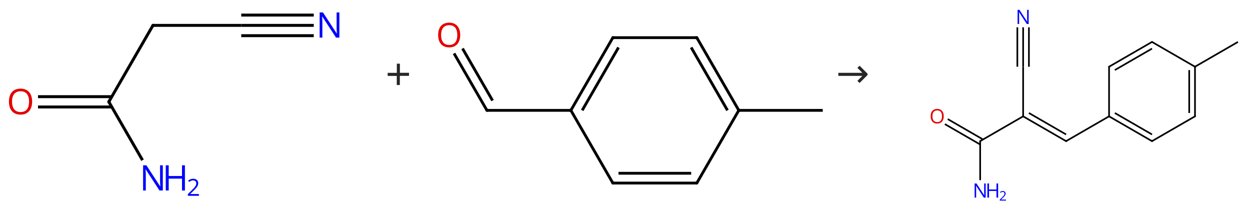
31-331-CAS-270680	Steps: **1** Yield: **98%**
1.1	**Catalysts:** 1-Octanesulfonic acid, 1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,8-heptadecafluoro-, ytterbium(3+) salt (3:1) **Solvents:** Toluene, Perfluorodecalin; 6 h, 80 °C
Experimental Protocols	

Ytterbium perfluorooctanesulfonate-catalyzed Knoevenagel condensation in a fluorous biphasic system

By: Yi, Wen-Bin; et al

Organic Preparations and Procedures International (2006), 39(1), 71-75.

Scheme 33 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (86) Suppliers (104) Suppliers (21)

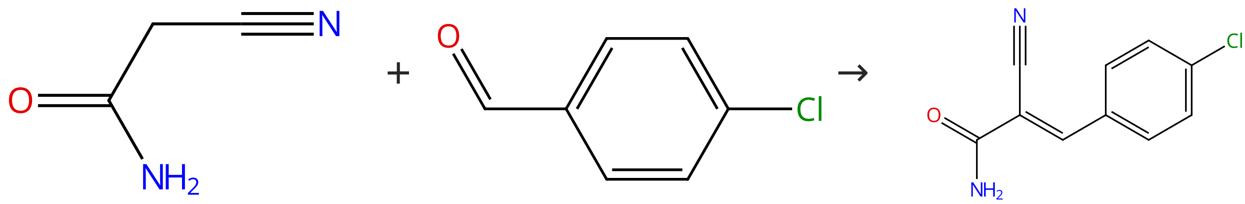
31-333-CAS-9244314	Steps: **1** Yield: **98%**
1.1	**Reagents:** Ammonium sulfamate; 20 s

Microwave-enhanced Knoevenagel condensation catalyzed by $NH_2SO_3NH_4$

By: Mogilaiah, K.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2010), 49B(3), 390-393.

Scheme 34 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (86) Suppliers (99) Suppliers (21)

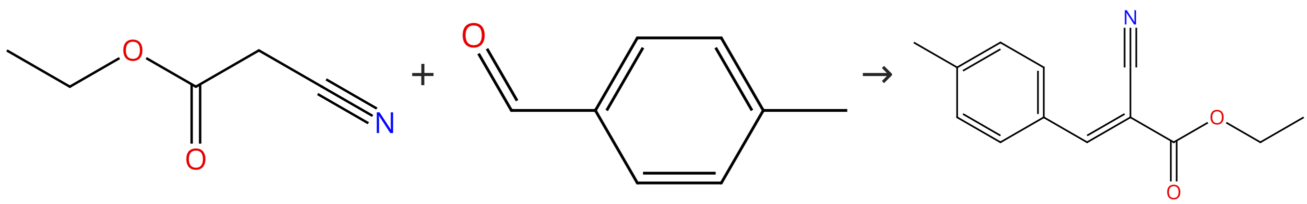
31-333-CAS-13501367	Steps: **1** Yield: **98%**
1.1	**Reagents:** Ammonium sulfamate; 10 s

Microwave-enhanced Knoevenagel condensation catalyzed by $NH_2SO_3NH_4$

By: Mogilaiah, K.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2010), 49B(3), 390-393.

Scheme 35 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (70) Suppliers (104) Suppliers (28)

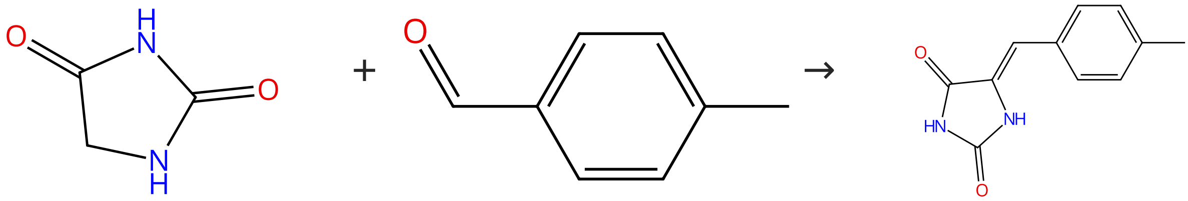
31-331-CAS-9580065	Steps: **1** Yield: **98%**
1.1 **Reagents:** Ammonium sulfamate; 8 s	

Microwave-enhanced Knoevenagel condensation catalyzed by $NH_2SO_3NH_4$

By: Mogilaiah, K.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2010), 49B(3), 390-393.

Scheme 36 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (85) Suppliers (104) Suppliers (5)

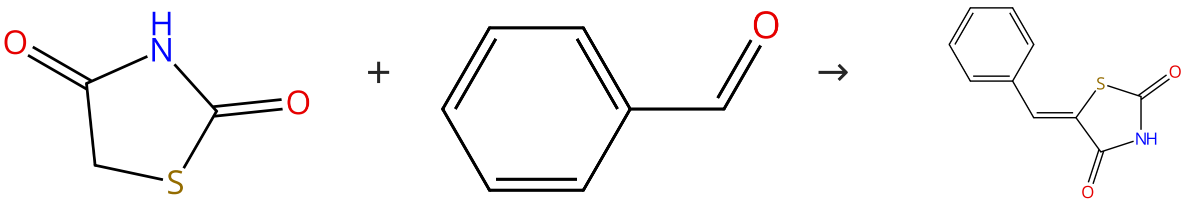
31-329-CAS-16001749	Steps: **1** Yield: **98%**
1.1 **Catalysts:** Ethanolamine **Solvents:** Water; 70 °C → 90 °C	
1.2 **Solvents:** Ethanol; 90 °C; 5 h, reflux	
Experimental Protocols	

Synthesis, biological evaluation and molecular modeling studies of psammaplin A and its analogs as potent histone deacetylases inhibitors and cytotoxic agents

By: Wen, Jiachen; et al

Bioorganic & Medicinal Chemistry Letters (2016), 26(17), 4372-4376.

Scheme 37 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (93) Suppliers (69) Suppliers (9)

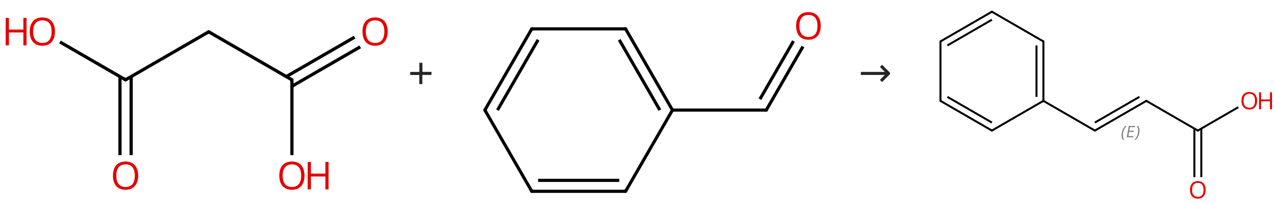
31-331-CAS-11347270	Steps: **1** Yield: **98%**
1.1 **Catalysts:** Urea; 5 min, 110 °C	

Urea/thiourea catalyzed, solvent-free synthesis of 5-arylidenethiazolidine-2,4-diones and 5-arylidene-2-thioxothiazolidin-4-ones

By: Shah, Sakshi; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(17), 5388-5391.

Scheme 38 (1 Reaction) Steps: **1** Yield: **98%**



Double bond geometry shown

Suppliers (87) Suppliers (69) Suppliers (159)

31-333-CAS-10149007	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Aluminum potassium sulfate dodecahydrate; 3 - 4 min, rt; 4 min, heated

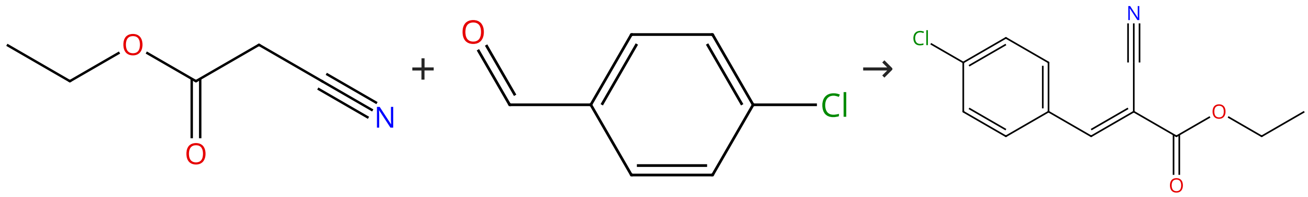
Alum [$KAl(SO_4)_2.12H_2O$]: an efficient, novel, clean catalyst for Doebner Knoevenagel reaction for the efficient production of α,β-unsaturated acids

By: Suresh; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2011), 50B(10), 1479-1483.

Scheme 39 (2 Reactions)

Steps: **1** Yield: **98%**



Suppliers (70)

Suppliers (99)

Suppliers (34)

31-331-CAS-13840047	Steps: **1** Yield: **98%**
1.1	**Reagents:** Ammonium sulfamate; 5 s

Microwave-enhanced Knoevenagel condensation catalyzed by $NH_2SO_3NH_4$

By: Mogilaiah, K.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2010), 49B(3), 390-393.

31-331-CAS-2020189	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Sodium chloride

Microwave assisted Knoevenagel condensation using NaCl and NH_4OAc-AcOH as catalysts under solvent-free conditions

By: Dave, Chaitanya G.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (2000), 39B(6), 403-405.